An Event-Triggered LLM Agent for LoRa Parameter Control over Hybrid Terrestrial–LEO Links: Doppler-Aware Simulation and Honest Baselines

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***Abstract*— Direct-to-satellite LoRa links change faster than adaptive data rate (ADR) can follow, and network events such as jamming or gateway outages are outside what ADR was designed for. We ask whether a small, local LLM agent, called only on events and acting through five validated tools, controls LoRa parameters better than ADR in a hybrid terrestrial–LEO network. We model LoRa under the Doppler rate of a 600 km pass at packet level with two receiver classes: with frequency tracking, our model loses SF12 near zenith below about 650 km, close to the 550 km measured on a CubeSat; without tracking, no spreading factor works over a whole pass. In an event-driven simulator with a class-A control plane and paired seeds, a rule-based controller on the same triggers as the agent gains +10.5, +15.0 and +12.9 points of delivery ratio over ADR during jamming, the end of a pass and a gateway outage. A 0.6B agent (Qwen3, 287 decisions, median latency 5.78 s on two CPU threads) never beat ADR and lost −17.5 points on average during a traffic surge, through actions that passed schema validation. Code, data and agent traces are public.**

***Index Terms*—** *LoRa, LEO satellite, direct-to-satellite IoT, Doppler rate, adaptive data rate, LLM agent, tool calling.*

I. Introduction

Direct-to-satellite IoT extends LoRa coverage to areas without terrestrial gateways by placing a gateway on a low-Earth-orbit (LEO) satellite [1]. The link is only available during passes of a few minutes, its SNR follows the elevation, and the carrier drifts by hundreds of hertz per second near zenith. Adaptive data rate (ADR) [2] was designed for static terrestrial links: it needs 20 uplinks (more than three hours at one reading every 10 minutes) and never raises the spreading factor (SF) on its own. Events such as jamming, blockage, gateway outages or the end of a satellite pass are outside what it was designed for.

Large language models (LLMs) that call tools [3] have been proposed for network operation [4]. This work asks a narrow question that can be measured: *can a small, local LLM agent, called only on events and acting through validated tools, choose LoRa parameters better than ADR and a classifier trained on oracle labels when the rules do not anticipate an event, and what does it cost in decision latency?*

Contributions. (i) A packet-level model of LoRa under a LEO Doppler rate with two receiver classes, validated against the exact noncoherent SER and compared with the 550 km limit measured on a CubeSat [5]. (ii) An open hybrid terrestrial/LEO LoRa network simulator with a class-A control plane and common random numbers across policies, so every comparison is paired. (iii) An event-triggered LLM agent with five guarded tools, evaluated against fixed SFs, ADR, an ML classifier, an oracle, and a rule-based controller that uses the same triggers and information. (iv) A measurement of the agent's cost (latency, calls, invalid actions) on a laptop-class CPU. All code, configurations, raw CSV files, agent traces and figure scripts are in the repository; every number in this report is generated from them.

II. System Model

A. Scenario

A rural cell holds N LoRa end devices that report a 20-byte reading about every 10 min (period uniform in $[540, 660]$ s, random phase). A terrestrial gateway (Okumura–Hata rural, $h_b = 30$ m, $h_m = 1.5$ m, log-normal shadowing $\sigma = 7$ dB per device plus 3 dB per packet) covers devices within 12 km; in the hybrid scenarios 20 % of the devices are placed 30–45 km away, outside terrestrial coverage, and rely on a LEO satellite gateway at 600 km ($G_{rx} = 6$ dBi, NF = 6 dB) that is usable above 10° of elevation. EU868: channels 868.1/868.3/868.5 MHz, 125 kHz, all in sub-band g1, hence a 1 % duty cycle per device (silent $99\,T_{oA}$ after each packet). Transmit power is 2–14 dBm and energy is $E = V\,I_{tx}(P)\,T_{oA}$ with the SX1276 RFO currents of the datasheet [6]; T_{oA} follows the Semtech formula [7].

B. LoRa physical layer under a Doppler rate

Symbol m of a chirp at SF is $x_m[n] = \exp\big(j2\pi(n^2/2N + (m/N - \frac{1}{2})n)\big)$, $N = 2^{SF}$, demodulated by dechirping and an FFT [8]. We validated the Monte Carlo SER against the exact noncoherent N-ary formula [9] (Fig. 1a). The static Doppler shift is removed using the preamble; what remains is the Doppler *rate* $\dot{f}_D$, which makes the residual offset grow linearly during the packet. Payload symbol k sees $f_k(t) = \dot{f}_D\,(t - t_{\mathrm{ref}})$, and an FFT bin is only $BW/2^{SF}$ wide (30.5 Hz at SF12). Two receivers bracket real hardware: **A** (*preamble*) corrects the offset once, at the preamble centre; **B** (*tracking*) runs a first-order decision-directed loop on the fractional peak position (Jacobsen estimator [10]) starting from the last preamble up-chirp, without knowing $\dot{f}_D$. Reduced-rate symbols (the 8-symbol header block, and the payload when DE = 1 at SF11–12) carry $SF - 2$ bits and tolerate ± 2 bins instead of ± 0.5. With CR 4/5 a single wrong symbol loses the packet, so the packet error rate is computed packet by packet rather

than with $1-(1-SER)^n$, which would ignore that drift concentrates errors at the end of the packet. The resulting tables $\mathrm{PER}(SF, \mathrm{SNR}, |\dot{f}_D|)$ (400 packets per point) are the only PHY input of the network simulator.

C. Satellite channel

Pass geometry is closed form for a circular orbit and a given maximum elevation (Earth rotation ignored): range $d(t)$, elevation, Doppler $f_D = -(f_c/c)\dot{d}$ and rate $-(f_c/c)\ddot{d}$. Path loss is free space plus the 3GPP TR 38.811 [11] suburban/rural S-band model (LOS probability, LOS/NLOS shadow fading, NLOS clutter loss of 16–19.5 dB), used as an approximation at 868 MHz.

D. Network and control plane

The event-driven simulator resolves collisions per gateway with pure ALOHA, as in [12] (same channel and SF, time overlap, 6 dB capture; SFs orthogonal). A packet succeeds if it survives collisions, its gateway is up, and a Bernoulli draw passes the PER table. Every policy-independent random number (positions, shadowing, reading times, per-packet fading, LOS state, the PER uniform, channel hopping) is pre-drawn per device and seed, so all policies see the same channel realisations and comparisons are paired. Network-side decisions reach a device only in the downlink after one of its delivered uplinks (LoRaWAN class A [13]); during a pass a broadcast satellite downlink can reach all devices. The network server only knows delivered uplinks, losses inferred from frame-counter gaps, the gateway noise floor and the satellite ephemeris.

III. Controllers

Fixed SF7 / SF12 use 14 dBm. **ADR** follows Semtech's recommended rule: after 20 uplinks, margin $= \max \mathrm{SNR} - \mathrm{SNR}_{\mathrm{req}}(SF) - 10\,\mathrm{dB}$, $N_{\mathrm{step}} = \lfloor \mathrm{margin}/3 \rfloor$, lowering SF then power, raising power if negative; it starts from its own fixed point on 20 earlier uplinks so that a 3 h run measures steady state. **ML-HGB** is a histogram gradient-boosting classifier that predicts (SF, power) after each delivered uplink from the last five SNRs, the device loss rate, current SF/power/link, satellite elevation, Doppler rate and an anomaly flag (the same information the agent can read); its labels are the oracle's choices for the device's next reading, collected on 8 training seeds disjoint from the 5 evaluation seeds. The **Oracle** sees the true channel of every packet (including its fading) and picks link, channel, SF and power with the lowest energy at PER $\leq 1\,\%$, or waits when no gateway can receive; it ignores collisions.

Event-triggered LLM agent. ADR runs by default. The agent is called only when the anomaly detector raises an alert, when the inferred loss rate exceeds 10 %, when the satellite rises, sets or will set within 2 min, and every 30 min (10 min cooldown per trigger). The user message only names the trigger category; details must be read with five tools: `get_link_stats`, `get_satellite_status`, `get_anomaly_report`, `get_network_load` and `set_tx_params(target, sf, tx_power_dbm, channels, link, duration_s)`. Targets are groups (all, terrestrial, satellite, affected, or device ids) and every change is a lease that expires back to ADR. Guardrails: schema and range validation (rejections are counted), "read before you write" enforced, at most 6 tool calls per decision, timeout, and fallback to ADR on any backend error. The system prompt states the role, LoRa facts computed from the code (thresholds, time on air, energy, FFT bin width), the objective (delivery first, then energy) and how actions take effect. Temperature is 0 with a fixed seed, and simulated time is frozen while the model thinks; wall-clock latency is recorded.

Table 1: Anomaly scenarios (event at $t = 1$ h, 3 h runs, $N = 300$).

S1 jamming	+15 dB on 868.1 MHz at the gateway, 30 min
S2 deep fade	+20 dB loss for 20 % of devices, 30 min
S3 pass end	70° pass ends at 1 h; 20 % remote devices
S4 gateway outage	terrestrial gateway down 30 min, 60° pass inside
S5 traffic surge	reporting rate ×5, 30 min

Triggered-Rules (not in the original plan) reacts to the same triggers with the same information and the same lease mechanism, using hand-written rules: avoid a jammed channel, raise SF of degraded devices, move devices to the satellite during an outage, make satellite devices wait outside passes, and spread SF7 devices over SF8 during a surge. It isolates the value of the LLM's reasoning from the value of the event-triggered architecture.

Detector. Rolling z-scores on the per-channel noise floor (every minute) and, per 5-min block, on per-channel mean SNR, the inferred loss rate (binomial test), the uplink count (Poisson test) and the share of silent devices, with two consecutive exceedances required; plus gateway silence over 5 min.

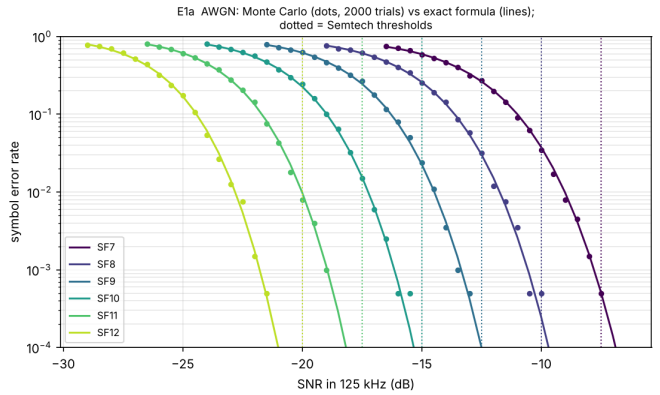
Protocol. Five scenarios (Table 1) start with an event at $t = 1$ h in a 3 h run with $N = 300$; every (scenario, policy) pair runs on the same 5 seeds. Metrics: PDR = delivered / sent, energy per delivered packet, recovery time (end of the last 10-min bin whose PDR is more than 5 % below the pre-event PDR, judged until 1 h after the end of the event; censored otherwise), and the invalid-action rate (rejected / total `set_tx_params` calls).

IV. Results

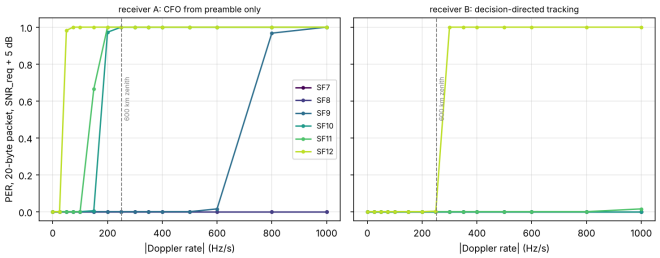
A. E1 – PHY validation

The Monte Carlo SER matches the exact noncoherent formula within 0.020 (absolute, 2000 trials per point, Fig. 1a). For a 20-byte packet, the ideal receiver reaches PER = 1 % between 0.32 and -1.40 dB from the Semtech thresholds (best margin at SF12: -1.4 dB), so the thresholds are consistent with the model without calibration.

The Doppler rate changes the picture (Fig. 1b). At $\mathrm{SNR}_{\mathrm{req}}$+5 dB, receiver A (no tracking) loses SF12 packets from 50.0 Hz/s, SF11 from 150.0 Hz/s and SF10 from 200.0 Hz/s: at the 600 km zenith rate the payload of an SF12 packet drifts by 9.23 bins. Receiver B (tracking) keeps SF7–SF11 error-free up to 1000 Hz/s and loses SF12 between 250 and 300.0 Hz/s, below the noise-free loop limit of 466 Hz/s.



(a)



(b)

Figure 1: E1. (a) SER in AWGN, Monte Carlo (dots) vs exact formula (lines). (b) PER of a 20-byte packet vs Doppler rate at $\mathrm{SNR_{req}}$+5 dB for both receivers.

B. E2 – Satellite pass

A 600 km zenith pass stays above 10° for 8.5 min, with Doppler up to ±19.7 kHz, a rate of 252 Hz/s at zenith, a one-way delay of 2.0–6.4 ms and a LOS SNR from -19.9 to -9.8 dB at 14 dBm (Fig. 2). The link only closes at SF≥9 and only near zenith for SF9–10: exactly where the Doppler rate is highest. With receiver A, SF12 is usable on 39 % of the pass (100 % if Doppler is ignored): *the most robust SF on the ground is the most fragile in orbit*, and no single SF is usable over the whole pass: the right SF changes with the elevation within minutes, far faster than ADR's 20-uplink history. With receiver B, SF12 only fails around zenith (usable on 98 % of the pass).

The altitude sweep (Fig. 3) is a check against measurements: with receiver B, SF12 is feasible at zenith from 650 km upwards (PER ≤ 1 %), while Doroshkin et al. measured SF12 failing near zenith below about 550 km [5], where our zenith rate is 279.2 Hz/s. Receiver B is therefore close to, and slightly more pessimistic than, real hardware; receiver A is far too pessimistic. Receiver B is used in all network experiments.

C. E3 – Scalability without anomalies

The simulator reproduces pure ALOHA ($|S - Ge^{-2G}| \leq 0.0026$ up to $G = 1.03$). In a terrestrial cell (Fig. 4), ADR delivers 97.5 % of the packets at $N = 100$ and 95.4 % at $N = 1000$, at 15.7 mJ per delivered packet. Fixed SF12 collapses with load (41.9 % at $N = 1000$) because each packet occupies the channel for 1.3 s; fixed SF7 stays near 84.8 % because distant devices

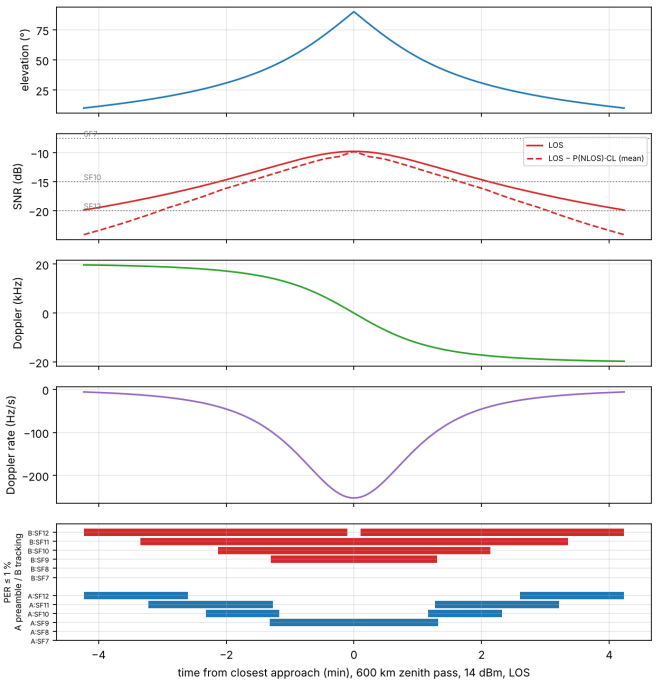


Figure 2: E2. 600 km zenith pass: elevation, SNR, Doppler, Doppler rate, and SFs with PER ≤ 1 % for receivers A and B.

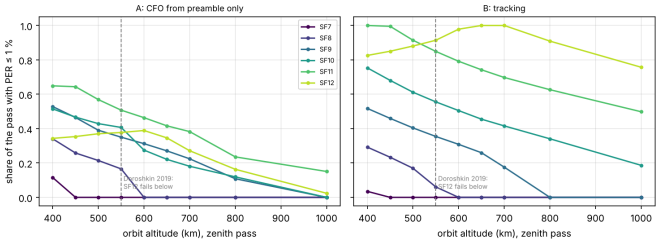


Figure 3: E2b. Share of a zenith pass where each SF reaches PER ≤ 1 %, vs orbit altitude.

cannot reach the gateway. The oracle wins at low load (98.1 %) but loses to ADR at $N = 1000$ (89.4 %): choosing the lowest energy per packet puts most devices on SF7 and increases same-SF collisions, while ADR's 10 dB margin spreads devices over SFs. The classifier trained on oracle labels under-provisions (75.4 % at $N = 1000$): it imitates an oracle that knows each packet's fading, and the most likely label given the observable features is a too-low power or SF. Triggered-Rules is identical to ADR here (no event, no action), which checks that the triggers stay quiet in normal conditions. The LLM agent equals ADR up to $N = 200$ but is below it in 7 of its 30 runs at higher load (worst seed −15.5 points at $N = 800$): each time, a false detector alert coincided with a periodic review and the model moved every device to SF12.

D. E4 – Anomaly scenarios

Fig. 5 shows the PDR over time and Table 2 the event-window PDR with the paired difference to ADR (same seeds, same

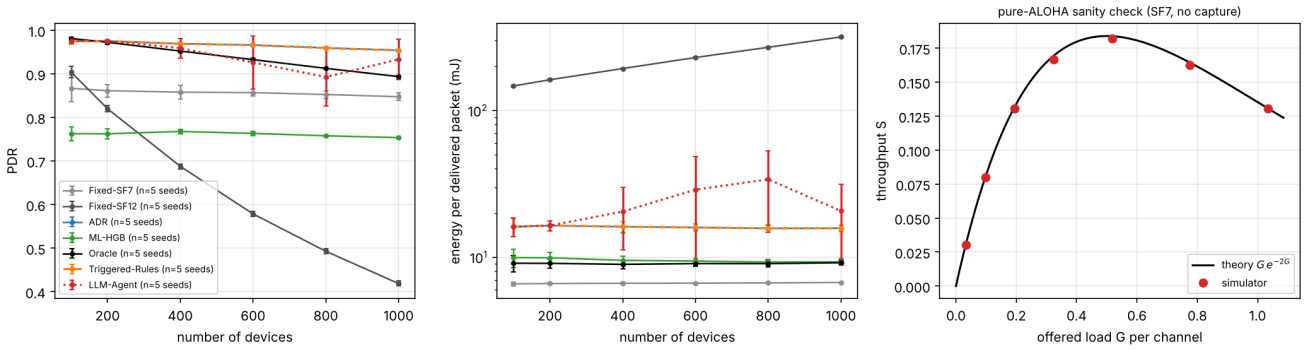


Figure 4: E3. PDR and energy per delivered packet vs number of devices (mean ± std, 5 seeds), and the pure-ALOHA check.

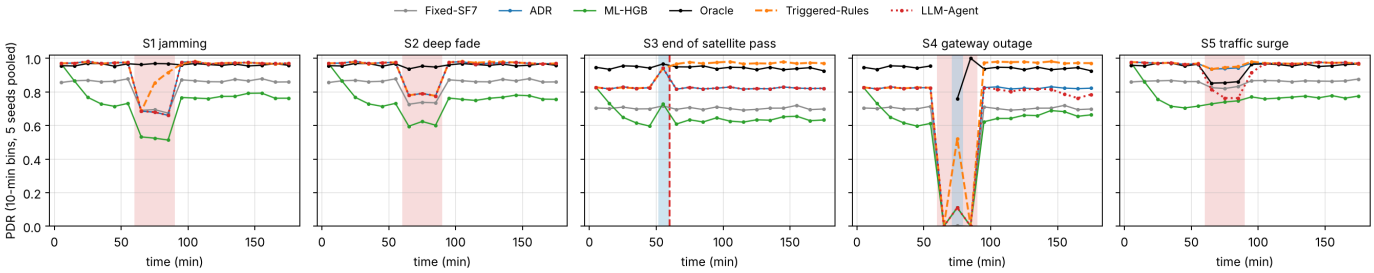


Figure 5: E4. PDR over time around the five events (10-min bins pooled over 5 seeds; red: event, blue: satellite pass). Fixed SF12 is off-scale and omitted. LLM agent: Qwen3-0.6B.

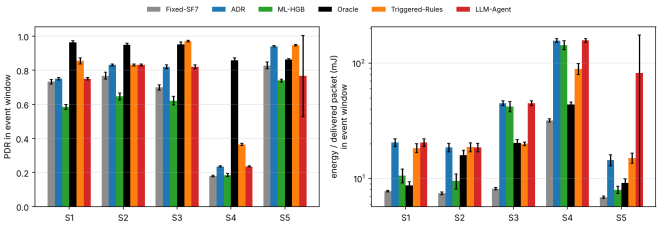


Figure 6: E4. Event-window PDR and energy per delivered packet (mean ± std, 5 seeds).

channel realisations).

ADR keeps the network healthy in normal conditions but does not react to events within their duration: during jamming its PDR falls to 75.1 % in the event window and only recovers when the jammer stops (30 min), because ADR can only raise power, never the SF, and decides on 20-uplink histories.

Triggered-Rules — the same triggers, information and lease mechanism as the agent, with hand-written rules — improves the event-window PDR by +10.5 points under jamming, +15.0 at the end of the pass and +12.9 during the gateway outage, on every seed, and halves the energy per delivered packet in S3 (19.9 vs 45.0 mJ). The event-triggered architecture is therefore worth about ten points by itself; this is the bar an LLM agent has to clear. No network-side controller helps against the deep fade (S2, +0.0 points): under class A a command needs a delivered uplink, and faded devices are silent. Only the oracle, which acts instantly and sees the fade, recovers (+11.9 points).

The *oracle* is the upper bound in S1–S4 but ends −7.8 points relative to ADR in the traffic surge: it minimises the energy of each packet in isolation, which puts most devices on SF7 and multiplies same-SF collisions. The *ML classifier* trained on its labels inherits this myopia and adds under-provisioning (−16.6 points in S1).

The LLM agent (Qwen3-0.6B [14], 25 runs, 287 decisions) read the network correctly with the tools but applied a change only 6 times; over E3 and E4, 15 of its 16 applied changes were the same action, every device to SF12 at 10 dBm. It is identical to ADR in S1–S3 (+0.0, +0.0 and +0.0 points). In S5 its event-window PDR was −17.5 ± 23.9 points relative to ADR, with a loss on 2 of 5 seeds: at the onset of the surge it moved every device to SF12 at 10 dBm for 30 min; an SF12 packet occupies the channel 23 times longer than an SF7 packet, the SF of most devices, while the traffic was five times higher. In one of the two runs its final message even stated that no change was required. The action was valid (in range, read-before-write respected), so the schema guardrail accepted it.

The detector, scored on the ADR runs, reaches an alert precision of 0.92 and a recall of 1.00; it flags jamming within 1 min and the outage within 5 min, but the deep fade only after 28 min, because faded devices fall silent instead of reporting a lower SNR.

Table 2: E4: event-window PDR (%, mean ± std over 5 seeds) and paired difference to ADR (points; seeds better / worse).

	S1 jam	S2 fade	S3 pass end	S4 outage	S5 surge
ADR	75.1±0.8	83.1±0.5	82.0±1.3	23.5±0.3	94.1±0.4
Fixed SF7	73.3	76.8	69.9	18.0	82.8
Fixed SF12	72.3	72.7	66.6	19.2	43.3
ML-HGB	58.4	64.6	62.1	18.4	74.0
Trig.-Rules	85.6	83.1	97.1	36.5	94.6
Δ	+10.5 (5/0)	+0.0 (0/0)	+15.0 (5/0)	+12.9 (5/0)	+0.5 (5/0)
LLM agent	75.1	83.1	82.0	23.5	76.6±23.9
Δ	+0.0 (0/0)	+0.0 (0/0)	+0.0 (0/0)	−0.1 (0/1)	−17.5 (0/2)
Oracle	96.5	95.0	95.1	85.8	86.3

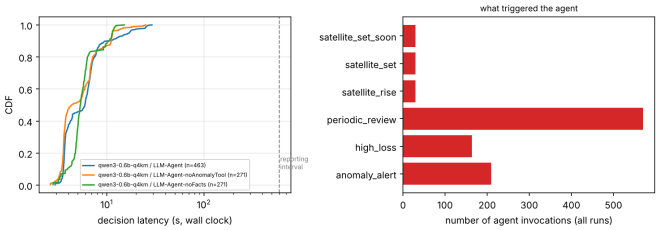


Figure 7: E5. Agent decision latency (CDF, wall clock, 2 CPU threads) and what triggered the agent.

E. E5 – Cost of the agent

On 2 CPU threads, a decision takes 5.78 s (median) and 14.67 s (95th percentile), with 1.22 tool calls and 7.65 LLM calls per simulated hour, most of them periodic reviews (Fig. 7). This is two orders of magnitude below the 10-min reporting interval: the reaction delay is set by class-A delivery (up to one reporting period), not by the model. 5 of the 11 `set_tx_params` calls were rejected (invalid-action rate 46 %), all for an empty target group; there was no read-before-write violation (0) and no backend failure.

F. E6 – Ablations

Removing the anomaly tool or the LoRa facts from the prompt made the 0.6B model entirely passive: over 271 and 271 decisions it applied 0 and 0 changes, read one tool per decision and stopped. Both variants are therefore identical to ADR in every scenario, including S5 (94.1 % and 94.1 % against 76.6 % for the full agent, Fig. 8). For this model, the context that makes it act is also what makes it harmful; an ablation can only show the value of a component if the full agent has some, which a larger model may provide.

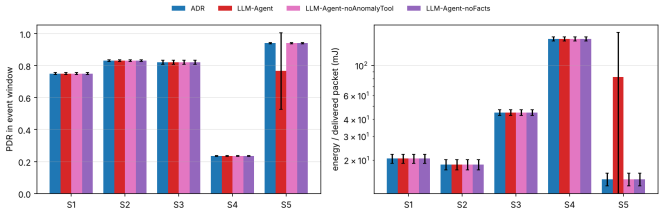


Figure 8: E6. Event-window PDR and energy per delivered packet for ADR, the full agent and the two ablations (Qwen3-0.6B, 5 seeds).

V. Discussion

Answer to the research question. With a 0.6B model, no: the agent did not beat ADR in any scenario, degraded the network in the traffic surge and under heavy load, and its few actions were almost always the same template rather than a response to the observed event. The event-triggered architecture, however, is clearly useful: the same triggers, telemetry and leased overrides driven by six hand-written rules gain 10–15 points in three of the five scenarios. An LLM agent is only worth its cost if it beats that rule set, for instance on combinations of events that no rule anticipates. Whether a 3B–7B model clears that bar is the next measurement; the scripts run unchanged with Ollama (`-backend ollama -model qwen2.5:3b`).

Validated is not safe. Range validation, the read-before-write rule and the tool cap all held, yet the agent's only recurrent action (every device to SF12) was harmful during a surge, and in one run its explanation contradicted its action. We tested a predicted-load check (reject an action whose offered load would exceed the ALOHA optimum $G = 0.5$) and found that it would not have blocked this case, because the surge had not been observed yet when the agent acted. Effective protection needs to act on consequences: roll an override back when the measured loss rises after it, apply it first to a subset of devices, or simulate it in a digital twin of the network before applying it — the simulator of this work could serve as that twin.

Physics first. The Doppler rate makes the best SF over a LEO pass change within minutes, much faster than ADR's 20-uplink history, and whether SF12 survives near zenith depends on the receiver's frequency tracking (Fig. 3). A controller that knows the ephemeris and the receiver's Doppler limits can choose the SF from the elevation; ADR cannot.

The control plane bounds every controller. Under class A, a command waits for the next delivered uplink. Agent latency (seconds) is negligible against this delay (minutes), and a deep fade cannot be fixed from the network at all; only device-side logic or class B/C downlinks could.

Oracles and imitation. A per-packet oracle that minimises energy is not an upper bound under heavy load, and a classifier that imitates it without its information under-provisions. Labels should target an expected PER given what the network can observe, and should account for collisions.

Limitations. No Earth rotation; S-band TR 38.811 tables at 868 MHz; orthogonal SFs; no background traffic at the satellite; a broadcast satellite downlink is assumed; one pass per scenario

and 3 h horizons; receivers A and B are models, not the Semtech demodulator; the agent results come from a 0.6B model run through llama.cpp because the development environment could not reach Ollama. All assumptions are listed in the repository (`docs/ASSUMPTIONS.md`).

VI. Conclusion

We built a reproducible, Doppler-aware simulation of a hybrid terrestrial–LEO LoRa network and evaluated an event-triggered LLM agent against fixed SFs, ADR, an ML classifier, an oracle and a rule-based controller on the same triggers. The physical layer reproduces the measured SF12 limit of LEO links within about 100 km of altitude. The event-triggered architecture gains about ten points of delivery ratio during jamming, satellite pass ends and gateway outages. A 0.6B agent did not exploit it and once caused harm that schema validation could not stop. Next steps: the 3B and 7B models, a consequence-aware safety layer, store-and-forward buffering, and several passes per day with Earth rotation.

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